LOCALITY ANNEALING AND THE FINITE INFORMATION HORIZON

A speculative framework connecting optoelectronic computation, quantum-inspired search, and the construction of local information

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Central proposal

A material or engineered optoelectronic system may support a process that progressively stabilises otherwise distributed information into locally addressable states. In this document, that proposed process is called *locality annealing*. A SHA-256 mining architecture is considered as a computational probe and constructor-like search process—not as evidence that a hash function can predict arbitrary data or create physical locality.

Status: hypothesis and research programme. The proposed material, universal limit, and 24-bit horizon are unverified. They are presented as conjectures to be operationalised and tested, not as established findings.

Abstract

This concept note proposes that locality can be treated as a constructible computational resource. A hypothetical special material, coupled to optoelectronic degrees of freedom, would enable a process termed *locality annealing*: the gradual reduction of a system's effective nonlocality until information becomes stably addressable within a bounded region, mode, or register. The proposal links this idea to a quantum-inspired SHA-256 mining system in which candidate headers are searched, evaluated, and accepted according to a hash predicate. The miner is interpreted as a constructor-like function over candidate states, while the hash itself remains a deterministic digest and does not disclose the input data.

A further conjecture is that the amount of information that can be locally constructed per optoelectronic qubit has a finite upper bound. The suggested figure—approximately 24 bits of accessible information beyond a chosen present-time reference—is treated here as a provisional scale-setting hypothesis. No known physical principle currently establishes this number as a universal limit. The document therefore defines measurable quantities, distinguishes computational from physical locality, and proposes experiments capable of supporting or falsifying the claims.

1. Motivation and conceptual framing

In ordinary computation, locality usually means that an operation depends on a restricted set of nearby variables or that information can be accessed through a bounded interaction. In physics, locality concerns the structure of interactions and the propagation of influence. These meanings are related but not interchangeable. A data structure can be computationally local without being spatially local, and a physical system can have local interactions while encoding globally correlated states.

The hypothesis begins with a stronger question: can a physical substrate actively *construct* locality? Rather than assuming that a useful local representation already exists, the system would anneal a distributed or ambiguous state into a stable local configuration. The term is intended to describe an experimentally observable transition, not to imply that space itself is literally manufactured by a computer.

2. The locality-annealing hypothesis

Let a system contain N addressable modes. Each mode has a state, a coupling pattern, and a measurable response to local interrogation. Define an operational locality score L(t) between 0 and 1, where larger values indicate that the state can be reconstructed or addressed using fewer, more spatially or structurally restricted interactions.

A simple phenomenological model is:

$$dL/dt = \alpha\, A(t) - \beta\, D(t) - \gamma\, L(t)$$

Here A(t) is the annealing drive, D(t) is measured disturbance or decoherence, and α, β, and γ are fitted parameters. This is a proposed measurement model, not a derived law. A candidate material would be interesting if it reproducibly increases L under controlled drive while maintaining energy, noise, and error budgets.

The hypothesised material would need to do more than exhibit ordinary optical switching or a reduction in circuit depth. It should produce a repeatable, quantifiable improvement in locality metrics compared with matched control substrates. Candidate mechanisms might include tunable coupling, phase-selective transport, nonlinear response, or dynamically reconfigurable optical pathways. These are possible mechanisms to investigate, not claims about a known special material.

3. SHA-256 mining as a constructor-like search

A SHA-256 miner evaluates candidate messages—often a fixed header combined with a nonce—until a digest satisfies a target condition. For a target requiring a specified number of leading zero bits, the expected number of independent trials is approximately 2^d, where d is the number of required leading zero bits, assuming hash outputs behave like uniformly distributed 256-bit values.

This search can be described abstractly as a constructor-like function: it generates candidate inputs, evaluates them, and returns one that satisfies a predicate. In that limited sense, the mining process constructs a witness to the target condition. It does not reconstruct arbitrary pre-existing data from the digest. SHA-256 is designed to make preimage recovery computationally infeasible; knowing a hash does not generally reveal the message that produced it.

A quantum implementation would not automatically make SHA-256 predictive. Quantum algorithms may offer speedups for particular search problems, but practical performance depends on the reversible circuit for SHA-256, error correction, hardware scale, and the precise task. Grover-style search gives a quadratic query advantage in an idealised unstructured search model; it does not provide a general method for reading arbitrary information from a hash.

The useful experimental question is therefore narrower: can an optoelectronic or quantum-inspired architecture organise candidate generation, comparison, and acceptance in a way that creates a measurable local computational advantage? A miner can be used as a demanding benchmark because it has an objective verification rule and a well-defined success condition.

4. What 'locality construction' would mean operationally

For the hypothesis to be testable, locality must be defined independently of the desired result. A practical protocol could encode a known message or candidate index across a set of optical modes, then measure how many local detector interactions are required to recover a designated portion of the state at a fixed error rate. The same task should be run on a conventional baseline and on the proposed annealing substrate.

Possible metrics include:

• **Local access cost:** number of detector reads, mode couplings, or communication steps required to recover a target bit string.
• **Reconstruction fidelity:** fraction of recovered bits that match the known input, measured over repeated trials.
• **Locality radius:** smallest interaction neighbourhood sufficient to reach a specified fidelity.
• **Annealing cost:** energy, time, optical exposures, and control operations required to reach the target locality score.
• **Stability:** persistence of the local representation under noise, drift, and repeated interrogation.

A claim of locality construction would require improvement in one or more of these metrics without quietly shifting the cost into an unmeasured channel. For example, reducing detector reads while increasing hidden preprocessing, calibration, or classical lookup would not establish a physical locality advantage.

5. The proposed finite information limit

The hypothesis suggests that there may be a universal upper bound on the complexity of locality construction per optoelectronic qubit. The proposed scale is approximately 24 bits of information beyond a selected present-moment reference. This is an intriguing conjecture, but the phrase '24 bits of universe' is not yet a defined physical quantity.

A bit count becomes meaningful only after specifying what is being counted. It could refer to the number of reliably distinguishable states, the size of a recoverable message, the conditional information gained from a measurement, or the number of future-labelled alternatives that can be resolved. These quantities are not equivalent. A qubit can carry a quantum state with continuous amplitudes, but the Holevo bound limits the classical information obtainable from a quantum system; for n qubits, no more than n classical bits can be reliably communicated per use under the usual assumptions. That familiar result does not imply a universal 24-bit limit.

To test a 24-bit proposal, define I_local as the number of reliably recoverable classical bits under a fixed protocol, error tolerance, energy budget, and observation time. Then test whether I_local per optoelectronic qubit approaches a stable ceiling near 24 as system size and repetitions increase. The result must be compared with standard information-theoretic bounds and with classical control or memory resources.

One possible operational definition is:

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I_local = log■(M_eff)
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where M_eff is the number of distinguishable local outcomes under a pre-registered error criterion. If M_eff is approximately 2²■ in a particular device, that would establish a device-specific capacity under the stated conditions—not a universal limit of the universe. A universal claim would require independent replications across materially different platforms and a theory explaining why the same bound appears.

6. 'Beyond the present moment': a careful interpretation

The phrase '24 bits beyond the present moment' can be retained as a motivating metaphor, but it should not be read as evidence that a machine has accessed future events or that the future universe has a measured finite size. In a scientific protocol, 'future' should refer to a prediction horizon: the set of future-labelled outcomes or states that the system attempts to distinguish before those outcomes are observed.

A valid prediction experiment must separate genuine forecasting from information leakage. The input data, labels, timestamps, and training records should be frozen before the prediction interval. The system should be tested on held-out outcomes, with baselines, blinded evaluation, and controls for cached information, deterministic seeds, and accidental access to future data. Success should be measured against a stated chance baseline and independently replicated.

7. Experimental programme

Phase A — Establish the computational baseline. Implement a conventional SHA-256 miner and independently verify every accepted digest. Record candidate rate, wall-clock time, energy, nonce range, and the exact header format. Separate hash-rate measurements from any claim about information recovery.

Phase B — Characterise the optoelectronic substrate. Measure mode coupling, detector noise, drift, cross-talk, and response under controlled optical drive. Use calibration patterns and randomised trials. Establish locality metrics before introducing the proposed annealing process.

Phase C — Test annealing. Apply the candidate annealing schedule and compare locality score, reconstruction fidelity, energy, and time against matched controls. Repeat across different input strings and optical configurations. Report failures and confidence intervals, not only successful runs.

Phase D — Test the information ceiling. Increase the number of modes and independent trials. Estimate I_local using a pre-registered distinguishability criterion. Examine whether capacity scales linearly, sublinearly, saturates, or changes with noise and coupling. A 24-bit ceiling should be accepted only if the data support it over plausible alternative models.

Phase E — Independent replication. Release circuit descriptions, source code, raw measurements, calibration procedures, and analysis scripts. Ask an independent group to reproduce the effect using a separate device and a separately generated test set.

8. Falsifiable predictions

The proposal would gain support if: (1) the candidate material produces a repeatable increase in independently measured locality score; (2) the improvement remains after accounting for preprocessing and control overhead; (3) the effect is robust across inputs and repeated trials; and (4) a reproducible information-capacity ceiling emerges under a clearly defined measurement protocol.

The proposal would be weakened or falsified if locality gains disappear under matched controls, if performance is explained by ordinary optical routing or classical caching, if the measured capacity continues scaling without a ceiling, or if the apparent 24-bit result depends on a post hoc choice of metric. A miner finding a valid nonce would verify the hash predicate, but would not by itself support locality construction or future prediction.

9. Current status and interpretation

At present, the special material, the locality-annealing mechanism, the universal per-qubit limit, and the 24-bit horizon remain unconfirmed hypotheses. The SHA-256 component is grounded in a known deterministic algorithm, but the proposed connection between mining and physical locality is an interpretation that requires independent measurement. Hash-rate anomalies, header breakdowns, or unexpected outputs should first be investigated as possible implementation, encoding, timing, or verification errors.

The strongest next step is not to infer a cosmic limit from a single computational result. It is to define the locality observable, instrument the optoelectronic system, verify the SHA-256 pipeline independently, and run blinded experiments that could prove the hypothesis wrong. A precise negative result would be scientifically useful; a reproducible positive result would justify a more formal theory.

10. Conclusion

Locality annealing is proposed as a way to describe the controlled conversion of distributed information into stable, locally addressable states. A SHA-256 mining architecture offers a verifiable search task that may serve as a benchmark for candidate-generation and acceptance processes. The more ambitious conjecture—that each optoelectronic qubit has a universal locality-construction capacity near 24 bits—requires a careful operational definition and substantial experimental evidence.

The central research question is therefore: *Can a material substrate produce a measurable, reproducible reduction in the physical or operational cost of local information access, and does that process exhibit a stable capacity limit?* Until tested, the 24-bit figure is a hypothesis, not a measurement of the universe.

Selected conceptual references

• C. E. Shannon, “A Mathematical Theory of Communication” (1948) — classical information and channel capacity.
• A. S. Holevo, “Bounds for the Quantity of Information Transmitted by a Quantum Communication Channel” (1973) — limits on accessible classical information from quantum states.
• L. K. Grover, “A Fast Quantum Mechanical Algorithm for Database Search” (1996) — quadratic query speedup for unstructured search.

• NIST, FIPS PUB 180-4, *Secure Hash Standard* — SHA-256 specification.

Document type: speculative research concept note. No experimental results are asserted in this document.